

Flow of SDLC, Waterfall Model & AI Integration Points

What we learn today

- What is SDLC
- Importance of SDLC in software projects
- Phases of SDLC (Requirement Analysis, Design, Implementation, Testing, Deployment, Maintenance)
- Flow of each phase with real-life examples
- Waterfall Model: Structure and Characteristics
- Pros and Cons of Waterfall
- Where AI assists in each SDLC phase
- AI-powered requirement analysis tools

Software Development Life Cycle (SDLC)



Requirement Analysis Phase

Requirement Analysis is the **first and one of the most important phases** of SDLC.

During this phase, the development team communicates with stakeholders to understand **what the customer actually wants**.

Activities

- ❖ Meet with clients and stakeholders
- ❖ Gather business requirements
- ❖ Identify user requirements
- ❖ Define functional requirements
- ❖ Define non-functional requirements
- ❖ Identify system constraints
- ❖ Prepare BRD (Business Requirement Document)
- ❖ Prepare SRS (Software Requirement Specification)
- ❖ Review and approve requirements

Documents

- ❖ BRD
- ❖ SRS
- ❖ Use Cases
- ❖ Requirement Traceability Matrix (RTM)

Design Phase

After requirements are approved, the solution is designed.

The Design Phase defines **how the system will be built**.

Developers don't start coding immediately, they first create the architecture and technical blueprint.

Activities

- ❖ High-Level Design (HLD)
- ❖ Low-Level Design (LLD)
- ❖ Database Design
- ❖ API Design
- ❖ UI/UX Design

HLD (High-Level Design)	LLD (Low-Level Design)
Focus: Overall System	Focus: Specific Module
Detail: High-Level Overview	Detail: Detailed Design
Prepared By: Solution/System Architect	Prepared By: Technical Lead/Senior Developer
Used By: Architect, PM, BA, QA	Used By: Developer, QA
Includes: Architecture, Modules	Includes: APIs, Logic, Database
Answers: <i>How is the system designed?</i>	Answers: <i>How is the module implemented?</i>

Implementation (Development) Phase

The Implementation Phase is where developers convert the approved design into actual software by writing source code.

Activities

- ❖ Coding
- ❖ Code Review
- ❖ Unit Testing
- ❖ Version Control (Git)
- ❖ Integration test
- ❖ CI/CD Integration

Testing Phase

Testing ensures that the software works according to the approved requirements and is free from critical defects before release.

Activities

- ❖ Test Planning
- ❖ Test Case Design
- ❖ Test Execution
- ❖ Defect Reporting
- ❖ Regression Testing
- ❖ Retesting
- ❖ Automation Testing
- ❖ Performance Testing
- ❖ Security Testing
- ❖ User Acceptance Testing (UAT)

Deployment Phase

After successful testing and approval, the application is deployed to the production environment where end users can access it.

Activities

- ❖ Production Release
- ❖ Database Migration
- ❖ Configuration Updates
- ❖ Smoke Testing
- ❖ Monitoring
- ❖ Rollback Planning

Maintenance Phase

Maintenance begins after deployment and continues throughout the software's life-cycle to keep the application functional, secure, and up to date.

Activities

- ❖ Bug Fixing
- ❖ Performance Improvement
- ❖ Feature Enhancement
- ❖ Security Patch Updates
- ❖ Customer Support
- ❖ Monitoring

Types of Maintenance

Corrective Maintenance

Fixes bugs found after release.

Adaptive Maintenance

Updates the software to work with new operating systems, hardware.

Perfective Maintenance

Adds new features or improves existing functionality based on user feedback.

Preventive Maintenance

Improves code quality and infrastructure to reduce future problems.

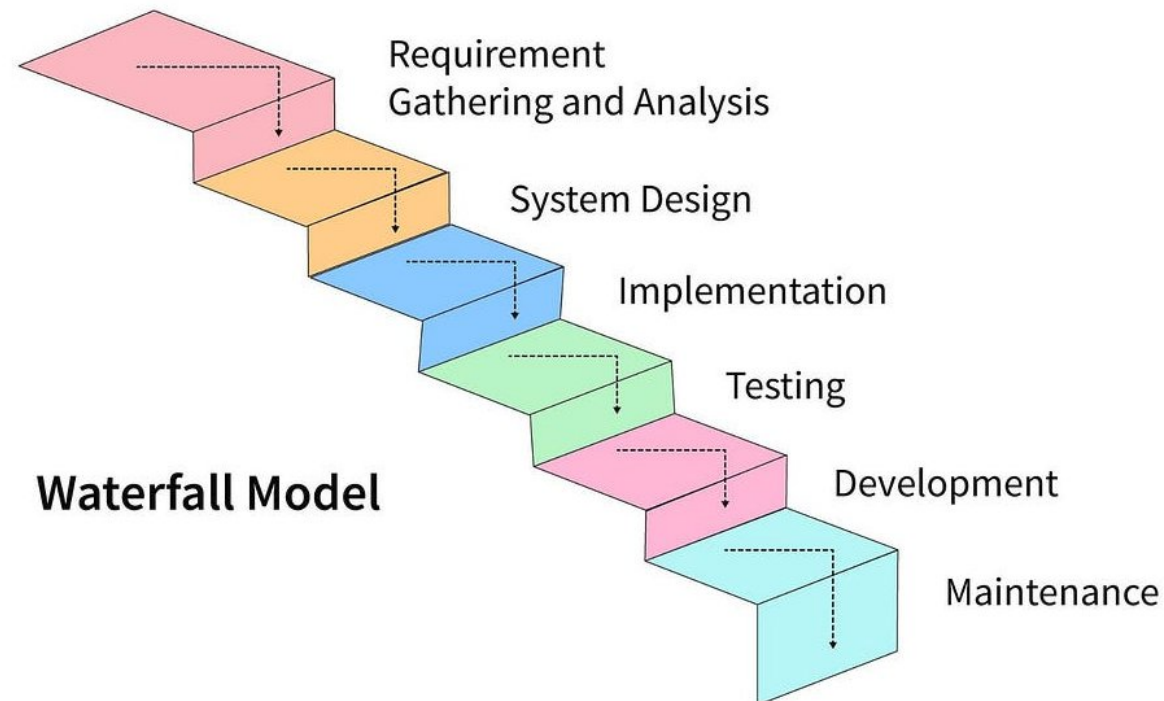
Real-Life SDLC Flow Example (Online Food Delivery App)

Phase	Example
Requirement Analysis	Customer wants food ordering, online payment, live order tracking
Design	Figma, database schema, API architecture
Implementation	Developers build the application
Testing	All kind of testing
Deployment	Software is published to the server
Maintenance	fix bugs, improve performance

What is the Waterfall Model?

The **Waterfall** Model is one of the oldest and most traditional Software Development Life Cycle (SDLC) models. It follows a linear and sequential approach, where the project progresses through a fixed sequence of phases.

Each phase must be completed, reviewed, and approved before moving to the next phase.



Pros and Cons of the Waterfall Model

Advantages (Pros)

- ❖ Simple and easy to understand.
- ❖ Follows a clear, sequential process.
- ❖ Well-defined phases and deliverables.
- ❖ Strong documentation throughout the project.
- ❖ Easy to manage and track progress.
- ❖ Suitable for projects with stable requirements.
- ❖ Supports regulatory compliance and audits.
- ❖ Easy knowledge transfer through documentation.
- ❖ Ideal for small to medium-sized projects with fixed scope.

Disadvantages (Cons)

- ❖ Difficult to accommodate changing requirements.
- ❖ Testing begins only after development is completed.
- ❖ Customer involvement is limited during development.
- ❖ High cost of fixing defects in later phases.
- ❖ No working software until the end of the project.
- ❖ Less flexible compared to Agile.
- ❖ Requirement mistakes are discovered late.
- ❖ Poor choice for rapidly changing or AI-driven projects.
- ❖ Longer delivery time for large projects.
- ❖ High project risk if initial requirements are incorrect.